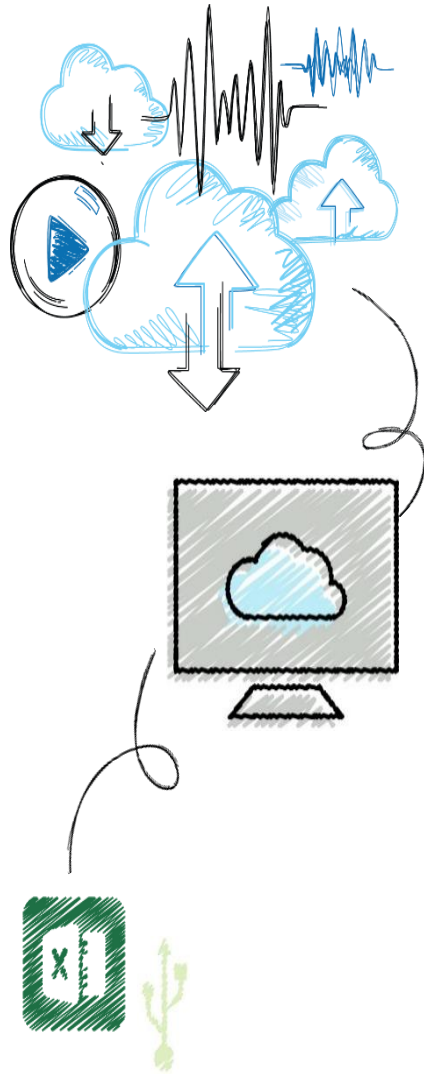


ICT703 Programming

Module 11

Python and SQL





Objectives

- Introduce SQL
- Database plus Application layer
- Python connection to MySQL
- Insert, update and query a database



Structured Query Language (SQL)

- SQL is a relational database system
- Records of data (rows) are stored in tables. Tables can join other tables to show the relationships.
- SQL is a high-level descriptive language – perfect for storing and retrieving data into a SQL Database.

PNo	FirstName	LastName	DOB	Employer
1	Fred	Smith	24/9/72	USC
2	John	Jones	1/1/84	QUT



- MySQL is a fast, easy to use relational database. It is currently the most popular open-source database
- MySQL is used for many small and big businesses. It is developed, marketed and supported by MySQL AB, a Swedish company. It is written in C and C++.
- MySQL is an open-source database, so you don't have to pay to use it.



Python with MySQL

- Python is good at applications, MySQL is good at database work
- Use python to get/manipulate data from people as well as display results.
- Python calls MySQL to store and retrieve data
- A good introduction to MySQL with exercises - www.w3schools.com/sql



Python with MySQL - Setup

- Firstly, we must install MySQL and then install the python package to have access to the functions to call MySQL.
- Download MySQL - <http://www.mysql.com/downloads/>
 - For this module we will not be using SQL Workbench but rather we will use a MySQL extension within VS Code.
 - Install the extension - MySQL v5.7.8 (Weijan Chen) - Database manager for MySQL/MariaDB, PostgreSQL, SQLite, Redis and ElasticSearch.
- MySQL Python Connector is used to access the MySQL database from Python, you need a database driver (install using the terminal window).

`py -m pip install mysql-connector-python`



MySQL – Build a table – We Need data

```
CREATE TABLE names (  
    ID INT AUTO_INCREMENT PRIMARY KEY,  
    first_name VARCHAR(255),  
    last_name VARCHAR(255),  
    dob DATE,  
    fav_colour VARCHAR(255),  
    chess_ranking INT  
);  
  
INSERT INTO names (first_name, last_name, dob, fav_colour, chess_ranking)  
values ('Andrew', 'Smith', '1976-12-12', 'Blue', 2000),  
('Jill', 'Smith', '1976-12-1', 'Purple', 1950),  
('Andrew', 'Jones', '1980-2-2', 'Pink', 1850),  
('Joanna', 'Jones', '1986-1-1', 'Blue', 2200);  
  
select * from names;
```



Python – Connect and retrieve

```
import mysql.connector #use this module

#connect to the database

mydb = mysql.connector.connect(host="localhost", user="root", password="root", database='name-db')

mycursor = mydb.cursor()

sql = "Select * from names"
mycursor.execute(sql)

myresult = mycursor.fetchall()

for x in myresult:
    print(x)
```



Python – MySQL – What's Next

- Practice!
 - Create tables in SQL
 - Use Python to insert into these tables
 - Use Python to update records
 - Use Python to display records.
- See the distinction between the database and the application layer